

**Katerina Bompai****(Attempt 1 of 1)****Test Information****QUESTION 1: TRUE/FALSE****5****out of 5 points**

The Gibbs Free Energy of an Ideal Gas is independent of Pressure



Given Answer: False

Correct Answer: False

QUESTION 2: CALCULATED NUMERIC**0****out of 5 points**The Heat Capacity of an Ideal Gas at $T=80^{\circ}\text{C}$ is equal to $C_p^{IG} = 89.314 \text{ kJ/(kmol K)}$. What is the value of $\left. \frac{\partial S}{\partial \ln T} \right|_V$ Express your result in **SI** units. **Round your answer to whole number.**

Given Answer: [None Given]

Correct Answer: 81

Answer range +/- 0.1 (80.9 - 81.1)

QUESTION 3: TRUE/FALSE**5****out of 5 points**

The Thermal Expansion Coefficient of an Ideal Gas is independent of Pressure



Given Answer: True

Correct Answer: True

QUESTION 4: CALCULATED NUMERIC

5

out of 5 points

Consider a gas that follows the equation of state $P(V - b) = RT$ where b is a constant. If the compressibility factor $Z = \frac{PV}{RT}$ is equal to $Z = 0.8$ at temperature $T = 45^\circ\text{C}$



what is the value of the residual enthalpy. Express your result in **kJ/kmol** and **round your answer to a whole number**

Given Answer: -529

Correct Answer: -529

Answer range +/- 5 (-534 - -524)

QUESTION 5: TRUE/FALSE

5

out of 5 points

The Heat of Vaporization of a fluid is a monotonically increasing function with respect to the temperature of the fluid



Given Answer: False

Correct Answer: False

QUESTION 6: CALCULATED NUMERIC

0

out of 5 points

What is the numerical value of $\left. \frac{\partial S}{\partial \ln P} \right|_T$ for an **Ideal Gas**.

Express your result in **SI** units. **Round your answer to two decimal digits.**



Given Answer: 1

Correct Answer: -8.31

Answer range +/- 0.01 (-8.32 - -8.30)

QUESTION 7: CALCULATED NUMERIC

0

out of 20 points

250 MMSCFD of Saturated steam, with quality equal to one, at $P_1=60$ bar expand adiabatically and reversibly through a steam turbine. If the power generated by the turbine is 50 MW, what is the pressure P_2 , of the steam at the outlet of the turbine.



Express your result in **kPa** and round your answer into **a whole number**.

Given Answer: [None Given]

Correct Answer: 30

Answer range +/- 0.5 (29.5 - 30.5)

QUESTION 8: CALCULATED NUMERIC

15

out of 15 points

Oxygen at $P_1=10$ bar and $T_1=25$ °C is compressed to $P_2=50$ bar in a single stage compressor. If the temperature of oxygen at the discharge(outlet) of the compressor is $T_2=230$ °C what is the efficiency of the compressor. Express your result as fraction (like 0.52). **Round your answer to two decimal digits**. Assume that during the compression Oxygen behaves like an Ideal Gas.



Given Answer: 0.81

Correct Answer: 0.81

Answer range +/- 0.01 (0.80 - 0.82)

QUESTION 9: CALCULATED NUMERIC

0

out of 20 points

Calculate the Internal Energy of Carbon Dioxide in **kJ/mol** at $T=20$ °C and $P=60$ bar. The reference state is Enthalpy $h= 200$ kJ/kg and Entropy $s=1$ kJ/(kg K) at $T_{ref}=0$ °C for saturated liquid.



Round your answer to a single decimal digit. Assume that in the vapor phase Carbon Dioxide behaves like an ideal gas.

Given Answer: [None Given]

Correct Answer: 12.6

Answer range +/- 0.1 (12.5 - 12.7)

QUESTION 10: CALCULATED NUMERIC

15

out of 15 points

Saturated liquid Propane at $T_1=50\text{ }^{\circ}\text{C}$, is throttled adiabatically to a pressure of $P_2=1\text{ bar}$. Calculate the vapor fraction of Propane after the throttling. **Round your answer to three decimal digits.** Assume that the vapor phase Propane behaves as an ideal gas.



Given Answer: ✓ 0.674

Correct Answer: ✓ 0.674

Answer range +/- 0.001 (0.673 - 0.675)

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